

Comprehensive Exams: Explicit Content Sources

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Subject Area 1: Systems Thinking (Dr. Wayne Wakeland)

1. Foundations of Systems Thinking

(a) Topic Selections

i. Concepts and Perspectives of Systems

A. Meadows and Wright, 2008 **[pp.1-203]**

B. Senge, 2006 **[pp. 163-187; 343-349]**

C. Lendaris, 1986

D. Bertalanffy, 1984 **[pp. 3-88]**)

E. **zwick_elements_2014** **[pp. 60-71])**

F. Ashby, 1955 **[pp. 86-117, 195-224])**

ii. Organization and Attributes of Systems

A. Simon, 1996 **[pp. 169-216])**

B. Barabási and Bonabeau, 2003

C. Barabási, 2009

D. Crutchfield et al., 1986

E. Mitchell, 2009 **[pp. 94-111; 169-185]**

F. Nicolis and Prigogine, 1989 **[pp. 5-44; 183-192; 217-242]**

G. Cover and Thomas, 2006 **[pp. 1-30; 34-35; 57-60]**

Subject Area 2: Systems Modeling (Dr. Wayne Wakeland)

1. Foundations of Systems Modeling

(a) Topic Selections

i. Model Typologies

A. Haefner, 2005 [pp. 3-15; 49-54]

B. Gilbert, 2005 [pp. 1-14]

C. Law, 2006 [pp. 1-6]

ii. Modeling Process: Formulation, Verification, Validation, Selection

A. Haefner (2005) [pp. 17-30; 144-175]

B. Gilbert (2005) [pp. 15-27]

C. Santner et al., 2003 [pp. 1-17; 199-203]

D. Law (2006) [pp. 243-273]

iii. Model Behavior: Sensitivity Analysis, Error Analysis, Stability Analysis, Chaos

A. Santner (2003) [pp. 189-199]

B. Haefner (2005) [pp.178-213]

C. Haefner (2005) [pp. 356-388]

D. May, 1977

E. May, 1976

iv. Experimental Design for Simulation Modeling

A. Santner (2003) [pp. 121-187]

2. Foundations of Model Archetypes

(a) Topic Selections

i. Qualitative Graphical Models

A. Ford, 1999 [pp. 69-95; 97-100]

B. Haefner (2005) [pp. 32-49]

ii. Probability Models: Bayesian DAGs; Markov Models; Hidden Markov Models; State Space Models

A. Hoff, 2009 [pp. 13-104; 125-207]

B. Lunn, 2013 [pp. 1-119; 137-177; 219-250]

C. Alpaydin, 2014 [pp. 387-484]

D. Zucchini and MacDonald, 2009 [pp. 3-74; 103-112; 135-235]

E. Fink, 2014 [pp. 107-123]

- F. Haefner (2005) [**pp. 215-231**]
- G. Shumway, 2011 [**pp. 319-378**]
- iii. Quantitative Compartment and Transport Models: Systems Dynamics and Time Series Regression Models
 - A. Luenberger, 1979 [**pp. 132-178; 224-248; 316-356**]
 - B. Lay, 2012 [**pp. 301-309**]
 - C. Haefner (2005) [**pp. 215-231**]
 - D. Gilbert (2005) [**pp. 28-45**]
 - E. Shumway (2011) [**pp. 47-153**]
- iv. Quantitative Particle Models: Agent-Based Models
 - A. Railsback and Grimm, 2012 [**pp. 35-45; 227-306**]
 - B. Gilbert, 2008
 - C. Resnick, 1994
 - D. Epstein and 2050 Project, 1996 [**pp. 153-178**]
 - E. Gilbert (2005) [**pp. 172-215**]
- v. Quantitative Stochastic Models: Discrete Systems (Queuing) Models
 - A. Kelton, 2010 [**pp. 202-228**]
 - B. Law (2006) [**pp. 6-32; 79-83; 275-353; 370-380; 471-477; 485-541; 669-671; 685-712**]

Subject Area 3: Ecology (Dr. Robert Scheller)

1. Fundamentals of Ecology

(a) Topic Selections

i. Biology

A. Curtis, 1983 [**pp. 49-75; 157-232**]

ii. Ecology

A. Curtis, 1983 [**pp. 951-993**]

B. Weier, 1982 [**pp. 347-371**]

C. Odum, 1969

D. Hairston et al., 1960

E. Cardinale et al., 2012

F. Estes et al., 2011

G. J. J. Wiens et al., 2010

iii. Evolution

A. Ehrlich and Raven, 1969

B. Alleaume-Benharira et al., 2006

C. Jay Gould and Eldredge, 1993

D. Kauffman, 1993 [**pp. 3-120; 173-286**]

E. Beinhocker, 2007 [**pp. 187-217**]

F. Feng et al., 2010

2. Foundations of Landscape Ecology

(a) Topic Selections

i. Historical Development

A. Solnetsev, 1948

B. Troll, 1950

C. Watt, 1947

D. J. A. Wiens, 2008

ii. Ecological Landscapes: Scales and Patterns

A. Turner et al., 1989

B. Wu, 2004

C. Legendre and Fortin, 1989

- D. J. A. Wiens, 1989
- iii. Ecological Landscapes: Dynamics and Processes
 - A. Turner, 2010
 - B. Turner, 1989
 - C. Peterjohn and Correll, 1984
 - D. Wegner and Merriam, 1979
 - E. Walker et al., 2010
 - F. J. A. Wiens, 1976
- iv. Measuring and Modeling Ecological Landscapes
 - A. Dale and Fortin, 2014 (**TBD**)
 - B. Naveh, 1987
 - C. Lefkovitch and Fahrig, 1985
 - D. Scheller and Mladenoff, 2007
 - E. Scheller, Robert M., 2013
 - F. Scheller et al., 2007

3. Foundations of Agroecology

(a) Topic Selections

- i. Concepts and Rationales
 - A. Tilman et al., 2002
 - B. Godfray et al., 2010
 - C. Gliessman, 2015 [**pp. 3-34; 301-344**]
 - D. Wolfe, 2000
 - E. Beus and Dunlap, 1990
 - F. Rosset, 2000
 - G. Thompson, 2010
- ii. Biotics and Abiotics
 - A. Schmidt et al., 2011
 - B. Gliessman, 2015 [**pp. 35-142**]
- iii. Agroecological Interactions
 - A. Gliessman, 2015 [**pp. 143-298**]
 - B. Gianinazzi and Schuepp, 1994
 - C. Schnitzer et al., 2011

- D. Smith et al., 2009
- E. van der Putten et al., 2013
- iv. Milpa Culture
 - A. Jiménez-Osornio et al., 1999
 - B. Toledo et al., 2008
 - C. Benjamin et al., 2001
 - D. Dahlin et al., 2005

Subject Area 4: Sustainability (Dr. Vivek Shandas)

1. Foundations of Sustainability

(a) Topic Selections

i. Fundamental Concepts

- A. Harris and Tufts University, 2001 [pp. xxvii-xxxvii]
- B. Marshall and Toffel, 2005

ii. Ecological Sustainability

- A. Pearce and Barbier, 2000 [pp. 102-129]
- B. Folke et al., 2004
- C. Schnaiberg et al., 2002

iii. Social Sustainability

- A. Sen, 2000[pp. 111-159]
- B. Pastor, 2007
- C. Miringoff et al., 1996
- D. Oecd, 2001
- E. Berbes-Blazquez et al., 2016

iv. Economic Sustainability

- A. Pearce and Barbier, 2000 [pp. 1-50; 157-209]
- B. Costanza and Daly, 1992
- C. Daly, 1974
- D. Goodwin, 2003

2. Foundations of Ecosystem Services

(a) Topic Selections

i. Ecosystem Services: Definitions and Metrics

- A. Millennium Ecosystem Assessment (Program), 2005 [pp.]
- B. Wallace, 2007
- C. Nelson et al., 2009
- D. Costanza et al., 1997
- E. Kremen, 2005
- F. Muradian et al., 2010
- G. Niemelä et al., 2010

- H. Norgaard, 2010
- ii. Global Ecosystem Health and Ecosystem Service Delivery
 - A. Zhao and Running, 2010
 - B. Hoegh-Guldberg and Bruno, 2010
 - C. Lobell et al., 2011
 - D. Dodds et al., 2013
 - E. Rogers, 2008
 - F. Eigenbrod et al., 2011

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